WORKED SOLUTION

STEP II 1990, Question 16

Probability and Statistics • Bayes' Theorem

STEP II 1990, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q16

The key idea

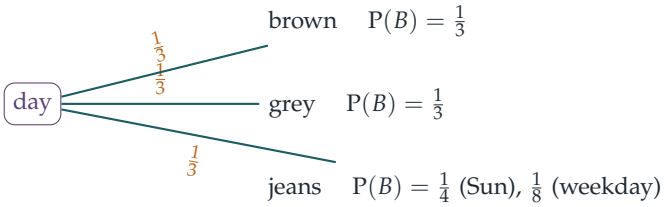
This is Bayes' theorem in disguise. The photograph tells us the black tie was worn, and we want the probability of the day *given* that. So we work forwards first: for each kind of day, find the probability of wearing the black tie, using a tree split by trousers. Then we turn the conditional probability round. The Monday rule only changes one branch of the tree, so the second part needs very little extra work.

Solution

Let S be the event that the photograph was taken on a Sunday, and B the event that I am wearing the black tie. The photograph is equally likely to come from any of the seven days, so before we look at it $\mathrm{P}(S) = \frac{1}{7}$. On every day except (later) Monday, each pair of trousers has probability $\frac{1}{3}$.

The chance of the black tie on each kind of day

- **Brown trousers.** The cravat is ruled out, so the tie is one of black, polka-dot or regimental, chosen at random: $\mathrm{P}(B) = \frac{1}{3}$.
- **Grey trousers.** The polka-dot tie is ruled out, so the choice is from black, regimental or cravat: $\mathrm{P}(B) = \frac{1}{3}$.
- **Jeans on Sunday.** A tie is always worn, picked from all four: $\mathrm{P}(B) = \frac{1}{4}$.
- **Jeans on a weekday.** With probability $\frac{1}{2}$ it's the sweat-shirt and no tie. Otherwise it's one of the four ties: $\mathrm{P}(B) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$.



Adding along the branches (the law of total probability over the trousers),

$$\mathrm{P}(B \mid \text{Sunday}) = \tfrac{1}{3}\left(\tfrac{1}{3}+\tfrac{1}{3}+\tfrac{1}{4}\right) = \tfrac{1}{3}\cdot\tfrac{11}{12} = \tfrac{11}{36} = \tfrac{22}{72},$$
$$\mathrm{P}(B \mid \text{weekday}) = \tfrac{1}{3}\left(\tfrac{1}{3}+\tfrac{1}{3}+\tfrac{1}{8}\right) = \tfrac{1}{3}\cdot\tfrac{19}{24} = \tfrac{19}{72}.$$

The first answer

By Bayes' theorem, with prior $\frac{1}{7}$ for Sunday and $\frac{6}{7}$ for the six weekdays together,

$$\mathrm{P}(S \mid B) = \frac{\frac{1}{7}\cdot\frac{22}{72}}{\frac{1}{7}\cdot\frac{22}{72}+\frac{6}{7}\cdot\frac{19}{72}} = \frac{22}{22+6\times 19} = \frac{22}{136} = \frac{11}{68},$$

as required. (The common factor $\frac{1}{7\times 72}$ cancels, which is why it's worth putting everything over 72.)

? **Ask yourself** Why isn't the answer just $\mathrm{P}(B \mid \text{Sunday})$, or $\frac{1}{7}$?

$\mathrm{P}(B \mid \text{Sunday}) = \frac{11}{36}$ answers the wrong question: it's the chance of the tie given the day, not the chance of the day given the tie. And $\frac{1}{7}$ ignores the evidence. The black tie is slightly more likely on a Sunday ($\frac{22}{72}$ against $\frac{19}{72}$), because there's no tieless option then, so the photograph nudges the probability of Sunday up from $\frac{1}{7} = \frac{11}{77}$ to $\frac{11}{68}$.

With the Monday rule

Now on Monday the jeans are certain, and Monday is still a weekday, so the sweat-shirt option still applies:

$$\mathrm{P}(B \mid \text{Monday}) = 1\cdot\tfrac{1}{2}\cdot\tfrac{1}{4} = \tfrac{1}{8} = \tfrac{9}{72}.$$

Sunday and the other five weekdays are unchanged. Each day still has prior probability $\frac{1}{7}$, which cancels, so

$$\mathrm{P}(S \mid B) = \frac{\frac{22}{72}}{\frac{22}{72}+\frac{9}{72}+5\cdot\frac{19}{72}} = \frac{22}{22+9+95} = \frac{22}{126} = \frac{11}{63}.$$

This is a little larger than $\frac{11}{68}$. That makes sense: on Monday the black tie becomes less likely ($\frac{1}{8}$ rather than $\frac{19}{72}$), so a black-tie photograph points away from Monday and slightly more towards Sunday.

Checks

- The tie probabilities for each day, together with "no tie", add to 1. For example, on a weekday the chance of no tie is $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.
- A simulation of 4×10^5 random days gives the proportion of black-tie days that are Sundays as about 0.161 (exact $\frac{11}{68} \approx 0.1618$), and about 0.174 with the Monday rule (exact $\frac{11}{63} \approx 0.1746$).

Common slips

- Forgetting that on Monday the jeans still come with the half-chance of the sweat-shirt: Monday is a weekday. Using $\frac{1}{4}$ instead of $\frac{1}{8}$ for Monday gives $\frac{22}{135}$.
- Choosing among all four ties with the brown or grey trousers. Each of those leaves only three ties.

Beyond STEP

Turning "probability of the evidence given the hypothesis" into "probability of the hypothesis given the evidence" is the heart of Bayesian inference. A neat way to say it: posterior odds equal prior odds times the likelihood ratio. Here the prior odds of Sunday against a weekday are $1 : 6$ and the likelihood ratio is $22 : 19$, which gives $22 : 114$, i.e. $11 : 57$. University courses build this up into Bayesian statistics, where whole distributions are updated in the same way. Blitzstein and Hwang, *Introduction to Probability*, has an excellent chapter on conditional probability with many examples like this one.

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